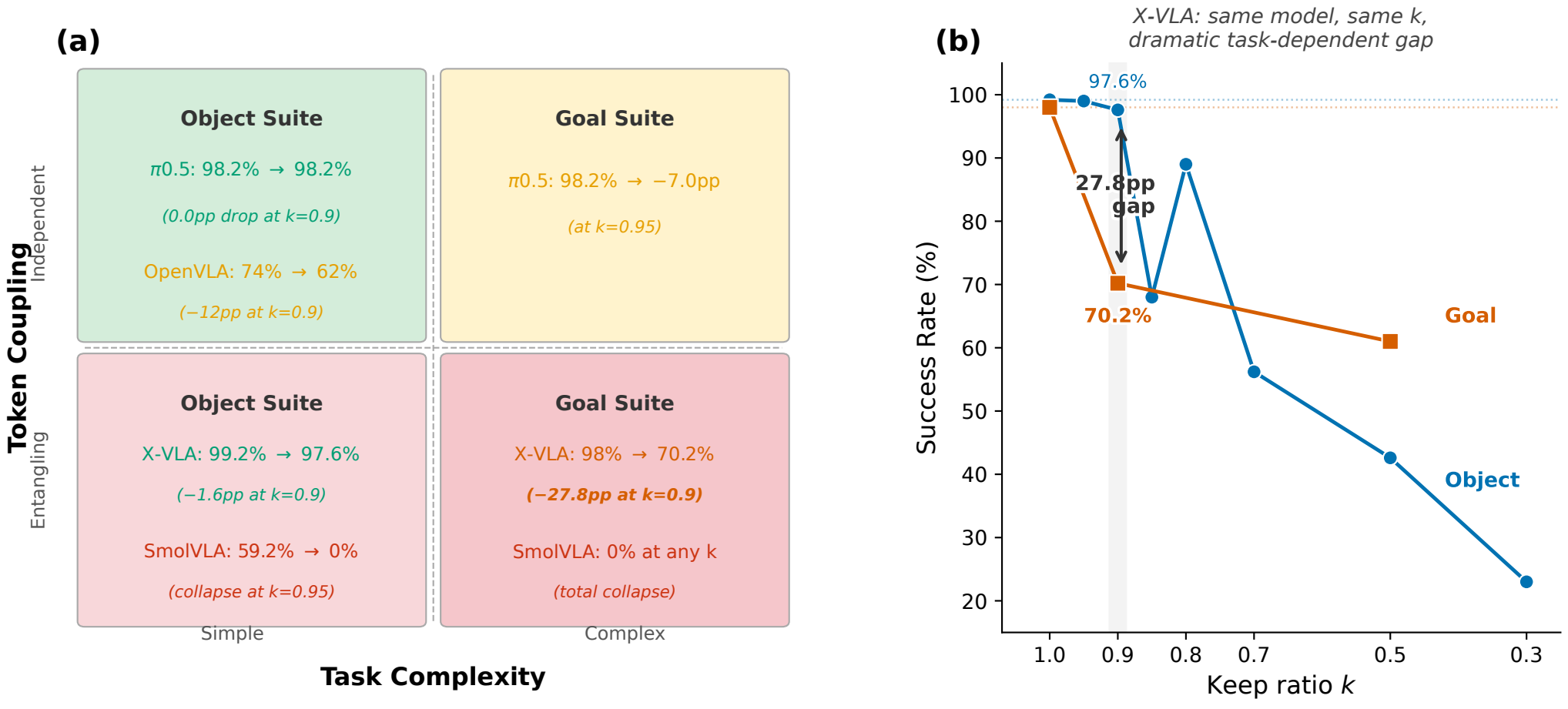
(a)
Object Suite
π0.5: 98.2% → 98.2%
(0.0pp drop at k=0.9)
OpenVLA: 74% → 62%
(−12pp at k=0.9)
Goal Suite
π0.5: 98.2% → −7.0pp
(at k=0.95)
Object Suite
X-VLA: 99.2% → 97.6%
(−1.6pp at k=0.9)
SmolVLA: 59.2% → 0%
(collapse at k=0.95)
Goal Suite
X-VLA: 98% → 70.2%
(−27.8pp at k=0.9)
SmolVLA: 0% at any k
(total collapse)
Token Coupling
Independent
Entangling
Simple
Complex
Task Complexity
(b)
X-VLA: same model, same k,
dramatic task-dependent gap
97.6%
27.8pp
gap
70.2%
Goal
Object
Success Rate (%)
100
90
80
70
60
50
40
30
20
1.0
0.9
0.8
0.7
0.5
0.3
Keep ratio k